

POST-LAB ASSIGNMENT

Detecting an ICMP (Ping) Flood

Snort 3 Intrusion Detection System Lab

Objective

The objective of this post-lab exercise was to independently configure Snort 3 to detect an ICMP (Ping) Flood against the OWASP Broken Web Apps (BWA) target, the test was conducted within the isolated VirtualBox laboratory network using Kali Linux as the traffic-generating host and OWASP BWA as the target.

Lab Environment

Component	Configuration
Source / Testing Host	Kali Linux — 192.168.56.1
Target	OWASP Broken Web Apps — 192.168.56.101
Virtual Network	VirtualBox Host-only network — vboxnet0
IDS	Snort 3
Packet Analyzer	Wireshark

Task 1: Write a Custom IDS Rule

A custom Snort rule was created to identify ICMP Echo Request traffic directed at the OWASP BWA target. ICMP Type 8 represents an Echo Request, which is the request generated by a standard IPv4 ping.

Custom Snort 3 Rule

```
alert icmp any any -> 192.168.56.101 any (msg:"ICMP Ping Flood Detected"; itype:8; sid:1000002; rev:1;)
```

The rule instructs Snort to generate an alert when ICMP Echo Request packets (Type 8) are observed from any source toward 192.168.56.101. The SID 1000002 provides a unique identifier for the custom rule and the revision number is set to 1.

Task 2: Launch the Attack

The ICMP flood was generated from the Kali Linux machine against the isolated OWASP BWA target using hping3. The --icmp option selected ICMP traffic, while --flood caused packets to be transmitted at a very high rate.

Attack Command

```
sudo hping3 --icmp --flood 192.168.56.101
```

```
Session Actions Edit View Help
~C
--- 192.168.56.101 hping statistic ---
40708488 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(kead@kali)-[~]
└─$ sudo hping3 -S -p 80 --flood 192.168.56.101
HPING 192.168.56.101 (vboxnet0 192.168.56.101): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
~C
--- 192.168.56.101 hping statistic ---
36743233 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(kead@kali)-[~]
└─$ sudo hping3 --icmp --flood 192.168.56.101
[sudo] password for kead:
HPING 192.168.56.101 (vboxnet0 192.168.56.101): icmp mode set, 28 headers + 0 data by
tes
hping in flood mode, no replies will be shown
~C
--- 192.168.56.101 hping statistic ---
39180630 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(kead@kali)-[~]
```

Figure 1. Kali terminal showing the ICMP flood command and hping3 flood-mode output.

Task 3: Traffic Analysis

Wireshark was used to observe the ICMP traffic in real time on the vboxnet0 interface. The display filter "icmp" was applied to remove unrelated traffic and focus the analysis on ICMP Echo Requests and Echo Replies between the Kali host and the OWASP BWA target.

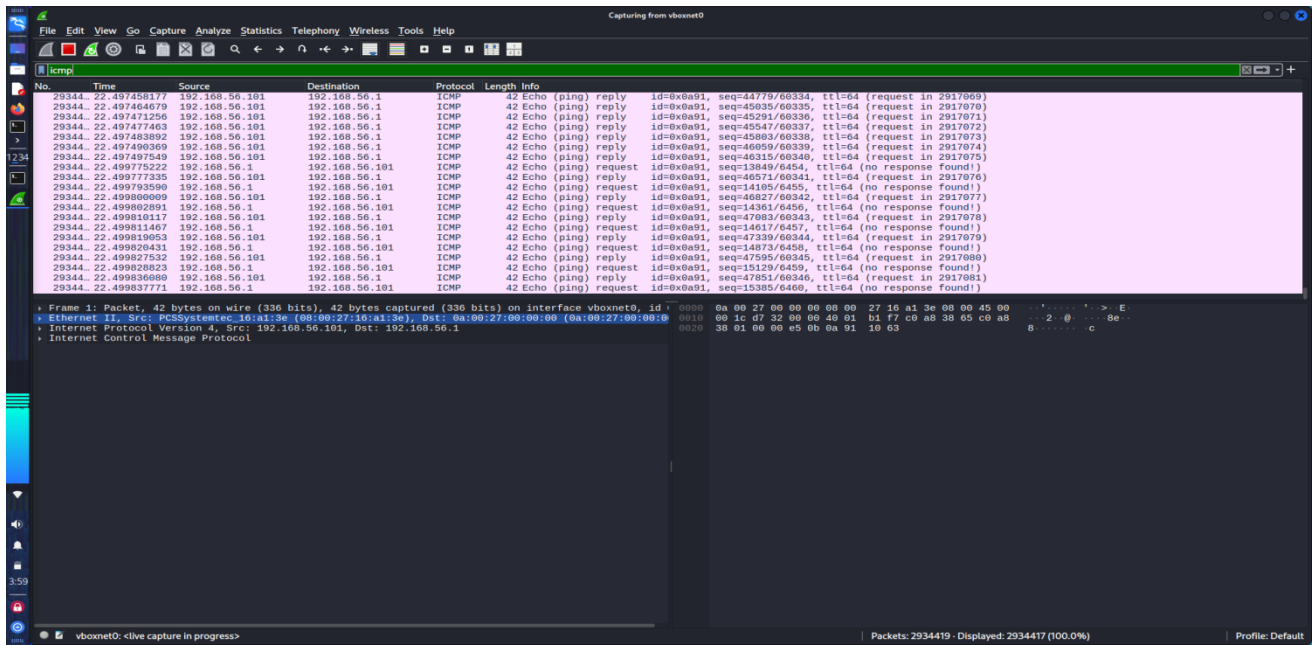


Figure 2. Wireshark capture filtered to ICMP traffic on vboxnet0.

IDS Detection Evidence

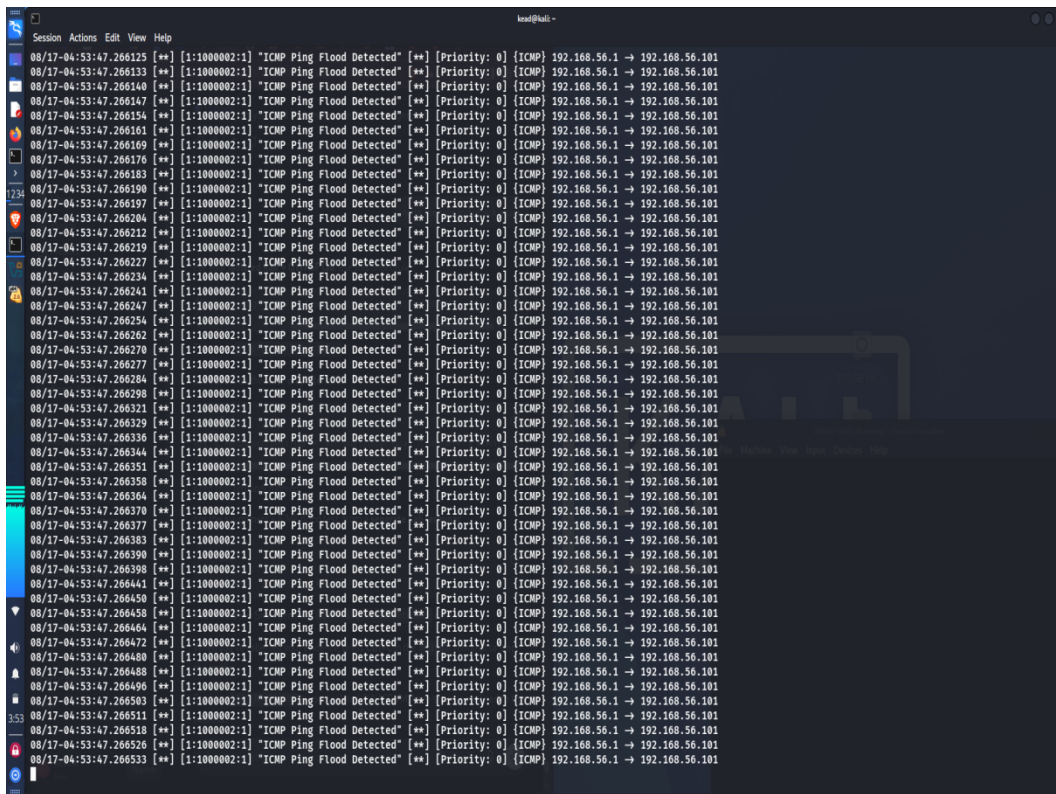


Figure 3. Snort IDS alert

Brief Analysis

An ICMP Ping Flood is a Denial-of-Service (DoS) technique in which a large volume of ICMP Echo Request packets is directed at a target in an attempt to consume network or system resources and reduce service availability. In this lab, Kali Linux generated the ICMP traffic toward the isolated OWASP BWA target, while Snort was configured to identify ICMP requests. Wireshark confirmed the presence of the ICMP Echo Request and Echo Reply traffic on the vboxnet0 interface. The exercise demonstrates how an IDS can identify suspicious ICMP traffic and provide an alert for further investigation.

Key Observation

The test was a DoS demonstration rather than a DDoS demonstration because the traffic originated from a single Kali host. A DDoS attack would involve multiple distributed sources generating traffic toward the same target.